



# Session 4: Aircraft Performance

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No. of Slides:

# Drift down Flight Planning & Depressurisation process & Escape route process

Airbus A320 Family

Boeing B787-9

# Aircraft Performance

## **Drift Down Flight Planning**

Drift Down is a maximum thrust/minimum rate descent necessitated by an engine failure in a multi-engine aircraft in the latter stages of climb or during cruise when an aircraft cannot maintain its current altitude and terrain clearance or other factors are critical.

## **EN-Route Limitations**

For Gulf Air EDTO operations, the following diversion scenarios are considered for planning purposes:

1. Engine failure;
2. Pressurization failure; and
3. Simultaneous engine failure and pressurization failure.

# Aircraft Performance

## **Engine Inoperative Diversion Policy**

Irrespective of the Single-Engine Speed used for EDTO planning purposes, the use of this particular speed schedule for an actual diversion is not mandatory. In the event of Engine failure, Gulf Air Policy is to initiate a descent as per the Obstacle Strategy in FCOM so as to minimize altitude loss while clearing the airway centerline and obtaining clearance.

Following this, the Commander shall complete an assessment of the following factors, and use his judgment to select a speed schedule appropriate for the circumstances applicable to the diversion:

1. Altitude capability of the aircraft, in relation to terrain clearance requirements along the proposed diversion route;

# Aircraft Performance

## **Engine Inoperative Diversion Policy**

2. The nature of the non-normal condition (e.g. cargo fire warning, engine failure), and structural integrity of the aircraft;
3. The actual or forecast weather along the proposed diversion route;
4. The actual fuel remaining, considering:
  - a. Higher speeds will minimize diversion time, but will result in higher fuel consumption and a lower level-off height.
  - b. Lower speeds will increase diversion time, but will result in lower fuel consumption and a higher level-off height.

# Aircraft Performance

## Engine Inoperative Diversion Policy

### Notes:

1. All EDTO routes are evaluated for obstacle clearance, based upon the net level-off altitude at the Single-Engine Speed declared by Gulf Air for the aircraft type concerned, and assuming maximum continuous thrust is set on the operating engine. Obstacle clearance is therefore not expected to be a limiting factor following an engine failure, provided the selected diversion speed is equal to or less than the declared Single-Engine Speed.
2. Commanders shall adhere to any specific diversion procedures applicable to an area, unless operational circumstances require a deviation from the procedure.

# Aircraft Performance

## **One-engine-inoperative Cruise Speed**

For Airbus/Boeing EDTO operations, Gulf Air has selected, and the BCAA has approved, a specific one-engine inoperative cruise speed.

1. B787-9: 0.84M/320KIAS
2. A330: 0.82M/320KIAS
3. A320 Family: 0.78M/320KIAS

The approved Single-Engine Speed is used for the following purposes:

1. Establishing the EDTO Area of Operation (the maximum diversion distance);
2. Establishing the EDTO diversion fuel requirements for the single-engine diversion;

# Aircraft Performance

## **One-engine-inoperative Cruise Speed**

3. Establishing the net level-off altitude to safely clear any enroute obstacle by the appropriate margin (unless a lower speed or the drift-down speed is required to clear the enroute obstacles); and 4. Conducting the diversion following an engine failure during an EDTO flight.

However, as permitted by the operational regulations, the Commander has the authority to deviate from the approved Single-Engine Speed, after completing an assessment of all the factors applicable to an actual enroute diversion.

# Aircraft Performance

## **One-engine-inoperative Cruise Speed**

In-flight the ONE ENGINE INOPERATIVE performance data is published in the Airbus FCOM (PER-OEI) and Boeing FCOM (Performance Inflight PI.63.1), along with:

1. Gross and net ceilings for the drift down (Airbus: green dot/Boeing: manoeuvre speed) and LRC speeds, with suitable corrections factors,
2. Summary of diversion speed strategies for each speed strategy: Airbus FCOM (PER-OEI) and Boeing FCOM PI.13.5.
3. Descent and cruise single-point performance data,
4. Diversion in-cruise quick check (fuel and time landing).
5. Emergency descent
6. Holding

# Aircraft Performance

## Aircraft Reference Gross Weight

ANTR OPS 1 requires the use of a specific aircraft reference gross weight, which must be used for the purposes of calculating and defining the EDTO Areas of Operation and the Maximum Diversion Distances.

| AIRCRAFT | REFERENCE GROSS WEIGHT |
|----------|------------------------|
| B787-9   | 240,000 KG             |
| A320     | 74,100 KG              |
| A321     | 89,000 KG              |
|          |                        |

# Aircraft Performance

## Maximum Diversion Distance and Area of Operation

The applicable maximum diversion distance from the adequate airports selected by Gulf Air to support the EDTO routes operated by each EDTO-approved fleet defines the EDTO Area of Operation.

The following maximum diversion distances are declared for each aircraft type:

| AIRCRAFT | Distance (NM)<br>for 60 minutes | Distance (NM)<br>for 120 minutes |
|----------|---------------------------------|----------------------------------|
| B787-9   | 443                             | 868                              |
| A320     | 405                             | 795                              |
| A321     | 395                             | 767                              |

### Notes:

1. These distances must not be used for determining if a route requires EDTO capability.
2. The 60-minute circles are used for non-EDTO flights.

# Aircraft Performance

## **EDTO Fuel Policy**

The two diversion scenarios that must be considered in determining fuel required for an EDTO flight are:

1. Fuel required after consideration of a diversion due to a simultaneous engine failure and pressurization failure occurring at the Critical Point/ ETP.
2. Fuel required after a consideration of a diversion due to pressurization failure only occurring at the Critical Point.

### **Notes:**

1. The fuel required for a single-engine diversion without pressurization failure is never limiting for fuel planning purposes.

# Aircraft Performance

## EDTO Fuel Policy

### Notes:

2. For the computation of the EDTO critical fuel reserves, the diversion fuel is defined as the total of:
  - a. Fuel burn-off from the ETP to the enroute alternate airport assuming a TAS equivalent to 320 KIAS (single-engine) or LRC (all-engines in case of depressurization) at 10,000 feet for the entire distance (i.e. no credit for drift-down is used for fuel calculation);
  - b. 15 minutes holding at 1500 ft. overhead the enroute alternate at the recommended holding speed;
  - c. Fuel for an initial approach (IFR) and missed approach, followed by a second approach (VFR) and landing (refer to Airbus FCOM PER and Boeing FCOM PD);

# Aircraft Performance

## EDTO Fuel Policy

### Notes:

d. The specific performance degradation factor for each individual aircraft, as stored in the computer flight planning system, will be applied automatically to the total burn-off from the ETP to the alternate airport (item 1 above) and holding fuel;

e. APU fuel consumption at the following rate:

B787-9: 120 kg/hr

A320 Family: 80 kg/hr

f. Contingency fuel, equivalent to 5% of the total burn-off for cruise, descent and APU consumption.

# Aircraft Performance

## EDTO Fuel Policy

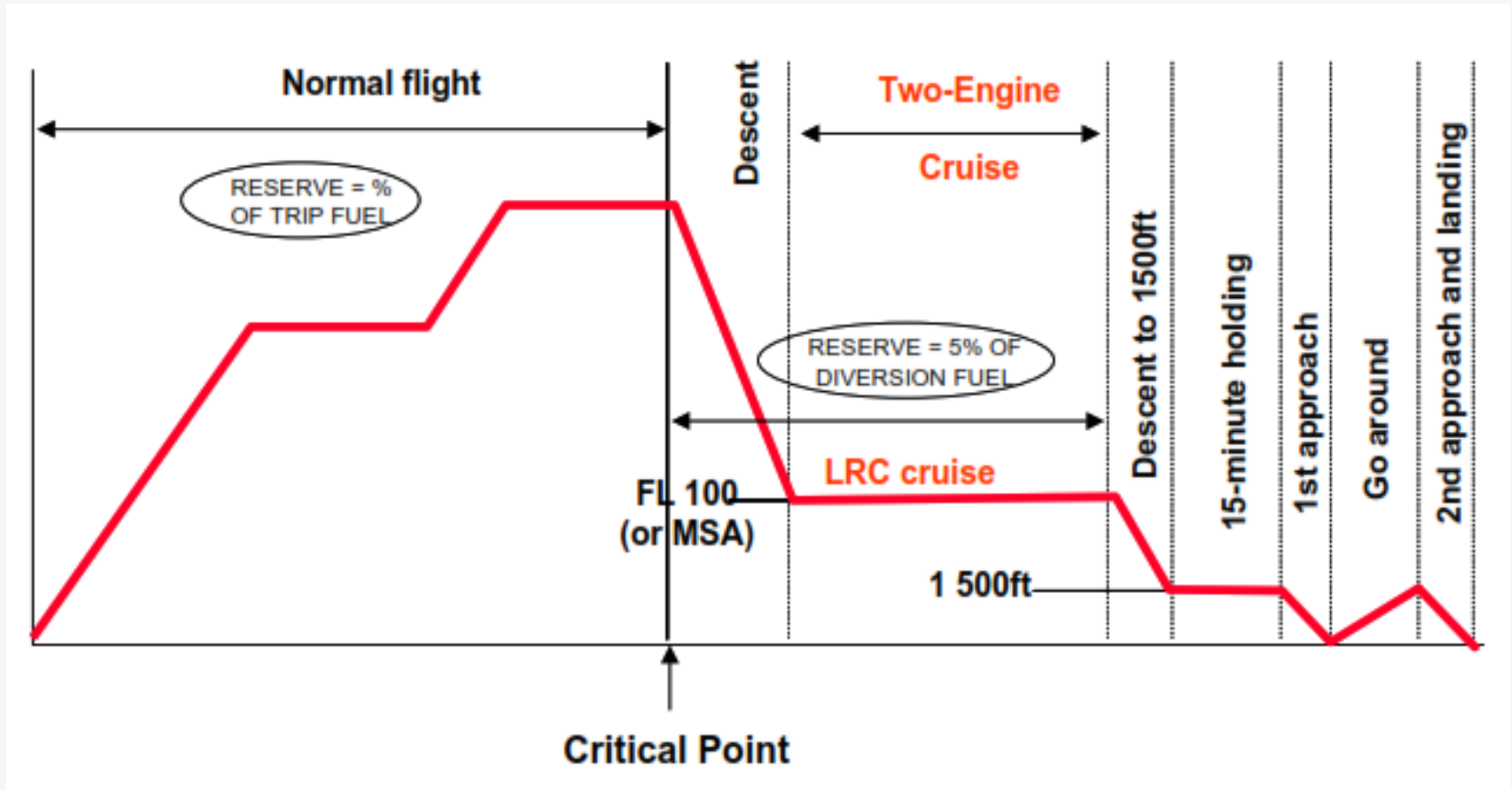
### Notes:

- g. As required, account will be taken for the effect on fuel consumption of any MEL and CDL item applicable to the particular aircraft
- h. Fuel to account for the following icing conditions:
  - a. 6% of the fuel required for the total diversion and holding period, to account for the use of wing de-ice/anti-ice systems.
  - b. A fuel allowance for additional drag, equivalent to 3% (single-and all-engines case – assuming one-hour exposure), of the total fuel required for cruise holding and missed approach, will be used to account for ice accumulation on unheated surfaces.

# Aircraft Performance

## EDTO Fuel Policy

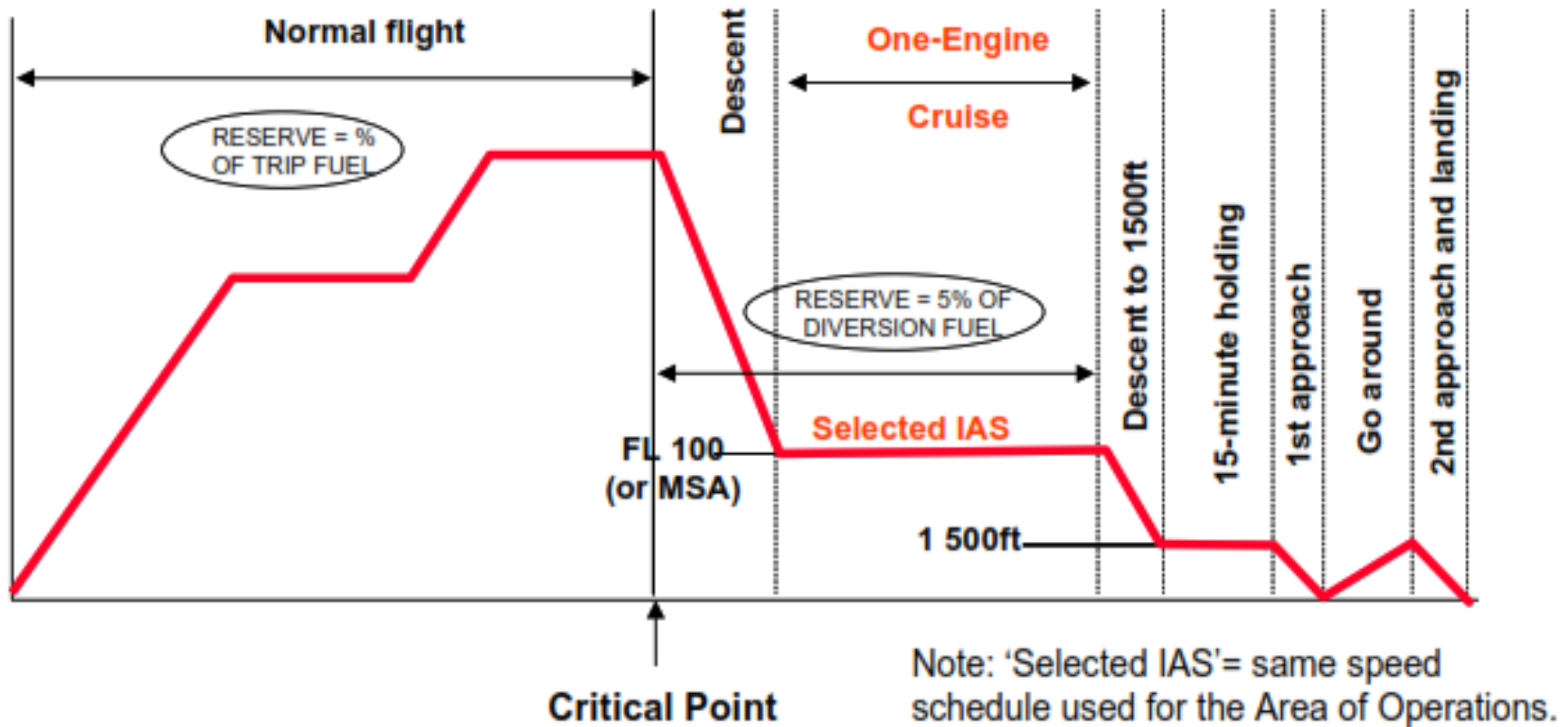
### Fuel for Depressurization (All Engines Running)



# Aircraft Performance

## EDTO Fuel Policy

### Fuel for Depressurization (One Engines out)



# Aircraft Performance

## **EDTO Fuel Policy**

### **Operational Flight Plan (OFP) - Fuel Analysis**

The fuel planning comparison specified above is automatically performed by the Operational Flight Plan (OFP) system, through an analysis process that compares at each ETP:

1. The fuel expected to be on-board,
2. The fuel required to divert to either one of the associated enroute alternate airports, considering the two failure scenarios defined above.

In the above analysis, the fuel expected to be on board when overflying an ETP considers the standard contingency fuel to be intact.

The Minimum Sector Fuel (MSF) derived by the OFP system ensures the greater of the two requirements above is met.

# Aircraft Performance

## ETDO and Terrain planning requirements in FPM

FPM allows calculation of flights according to EDTO, ETP and Terrain planning requirements. The EROPS page offers options for and results of the EROPS and Terrain Clearance calculations. For details refer to FPM user guide in shear point

FWZ4040-1,VIE-BGI,29MAY12,VAR 1 - FltPrep

List | Flight | Variants | Details | EROPS | MEL | Comm | Met | Notam | Monitor | Info | Errors

Quick EROPS  Ignore suitability check for:

EGKK EGKA LFRK LFRJ LFBG LFOV LFBG LFOU LFDL  
LFRK EGHJ LFXB LFDN LFRN LFRJ LFCY LFBH EGHJ LFRS LFBG  
LFRD EGJJ EGJA LFER LFRZ EGJB LFBG LFRJ LFRJ LFRV  
LFRD LFRH LFRU EGHJ LFRQ LFRB EGHJ LEAS LECO LEST LPPD  
LPAZ LPLA LPGR LPSJ LPPJ LPHR LPFL TFFF TBPB TLPC TLPL  
TVSV TTCP TGPY TTTP

Mode: ☒ Auto ☐ Manual

Wind / Ice: ☒ Actual wind file ☐ Average WC

EROPS: ☒ Eng. Anti Ice ☐ E+A Anti Ice ☐ Ice Accr.

Terrain: ☐ Pad Alt  Pad GM

Diversion Times: IHT: 60 MDT: 180 Max MDT: 180

| ID | EAlt | WC | ESAD | FL  | MinF | TripF | TripT | MinTT | MaxTT | ETO  | From | To    | ADDF  |       |
|----|------|----|------|-----|------|-------|-------|-------|-------|------|------|-------|-------|-------|
| 1  | 060  | 3  | LECO | T25 | 378  | 100   | 9012  | 7601  | 1.05  | 1.05 | 1.16 | 13.27 | 14.32 | 15.14 |
| 2  | PET  | 4  | LECO | T28 | 463  | 100   | 13187 | 9501  | 1.19  | 1.19 | 1.32 | 13.42 | 14.32 | 15.14 |
| 3  |      | 4  | LPPD | H35 | 463  | 100   | 13279 | 9589  | 1.19  | 1.19 | 1.33 | 13.42 | 15.01 | 15.15 |
| 4  | 060  | 5  | LPPD | H34 | 438  | 100   | 10226 | 8815  | 1.15  | 1.15 | 1.28 | 13.46 | 15.01 | 15.15 |
| 5  | 060  | 7  | LPFL | T36 | 367  | 100   | 8835  | 7424  | 1.04  | 1.04 | 1.14 | 15.52 | 16.56 | 17.06 |
| 6  |      | 7  | LPLA | T39 | 474  | 100   | 13574 | 9870  | 1.21  | 1.21 | 1.35 | 15.52 | 17.13 | 21.03 |
| 7  | PET  | 8  | LPLA | T30 | 1118 | 100   | 27412 | 23049 | 3.05  | 3.05 | 3.43 | 17.20 | 17.13 | 21.03 |
| 8  |      | 8  | TBPB | H4  | 1118 | 100   | 27670 | 23295 | 3.05  | 3.05 | 3.43 | 17.20 | 19.57 | 21.03 |
| 9  | 060  | 9  | TBPB | T3  | 397  | 100   | 9558  | 8147  | 1.08  | 1.08 | 1.20 | 18.49 | 19.57 | 21.03 |

Airports used: IA:LOWW-LFEY-LECO-EA:LECO-LPPD-IA:LPPD-LPFL-EA:LPLA-TBPB-IA:TBPB

MDT/TAS opt: 180 / 400

Ready Standard M

Routing: LOWW 0000 0 WW304 0000 3 WW293 0000 8 WW232 0000 13 WW177 0000 44 LUGIM 0000 65 LNZ 0000 105 SBG 0000 156 (\*\*\*) 0000 157 TRAUN 0000 168 ROTIN 0000 182 MANAL 0000 200 KONIN 0000 218

Range:   ☐ Suitable only

Airport:  Distance:

# Escape Route process

Airbus A320 Family

Boeing B787-9

# Aircraft Performance

## **Methods to Establish Escape Route Charts (Depressurization/Drift-down)**

Escape Route charts are produced by Gulf Air to assist Crew in areas of significant terrain, where descent to 10,000ft may not be immediately possible and may lead to a more complex navigation manoeuvre, as it is not easy to determine the critical point and/or routing to safe height/enroute alternate by using the Jeppesen Enroute Charts.

The published flight altitudes are calculated using the following assumptions:

1. Descent schedules as per Aircraft Manuals with gear up at Mmo / Vmo, structural considerations permitting.
2. Enroute descent between safety heights at 2,000 ft./min.
3. Two-minute allowance has been included for turns when calculating the maximum distance possible. If a turn of 180° is required, re-establish on track as soon as possible.

# Aircraft Performance

## **Methods to Establish Escape Route Charts (Depressurization/Drift-down)**

4. Maximum passengers.
5. Depressurization occurs at the critical point (i.e.: Decision Point).
6. Conservative speeds of:
  - a. 400 kts TAS for B787-9,
  - b. 400 kts TAS for A330, and
  - c. 390 kts TAS for A320 Family.
7. Complete exhaustion of the passenger oxygen system at approximately:
  - a. 22 min – A320 Family, A330 and B787-9 with descent to 14,000 ft. or below within this time period. Further flight above 10,000 ft. but not exceeding 14,000 ft. is permitted for a period not exceeding 30 min. After this time, supplementary oxygen must be available for a minimum 10% of total passengers on board for the duration of flight at these levels.

# Aircraft Performance

## **Methods to Establish Escape Route Charts (Depressurization/Drift-down)**

8. Diversion may be made to any suitable airport along the diversion route, subject to prevailing conditions.
9. A minimum RFF Category of 4
10. In case of no VORs reception at low altitudes, the GPS positioning can be utilized for navigation.
11. GPS primary is available; the navigation aid or waypoint shall be entered in the progress page.

# Aircraft Performance

## **Methods to Establish Escape Route Charts (Depressurization/Drift-down)**

### **Obstacle Clearance Policy**

Obstacle clearance is applicable for flying on airways, flying off airways and flying direct segments.

Having consideration for all obstacles within a 15 NM buffer on each side of the charted track and 15 NM radius at end fixes, except when diverting to an airport where the MSA covers 15 NM either side of the preferred arrival route.

Obstacle data is usually extracted using the Grid MORA data published by Jeppesen. However, in areas where the Grid MORA is limiting, more precise obstacle data is extracted from topographic charts.

Published altitudes provide a 2,000 ft. buffer plus the topographic chart tolerance, rounded up to the next 1000ft.

# Aircraft Performance

## **Methods to Establish Escape Route Charts (Depressurization/Drift-down)**

## **Enroute Performance Requirements - All Company Aircraft - One Engine Inoperative**

ANTR OPS 1.500

The mass of the airplane at the pre-flight or in-flight re-planning stage shall be such that:

1. the one engine inoperative enroute net flight path has a positive gradient at 1500ft above the aerodrome where the landing is assumed to be made after engine failure; and

# Aircraft Performance

## **Methods to Establish Escape Route Charts (Depressurization/Drift-down)**

### **Enroute Performance Requirements - All Company Aircraft - One Engine Inoperative**

ANTR OPS 1.500

2. the gradient of the one engine inoperative net flight path must be positive at least 1000ft above all terrain and obstructions along the route within 5 NM on either side of the intended track; or
3. the one engine inoperative net flight path must permit the airplane to continue flight from the cruising altitude to an aerodrome where a landing can be made while clearing vertically, by at least 2000 ft., all terrain and obstructions along the route within 5 NM on either side of the intended track.

# Aircraft Performance

## DECOMPRESSION PROCEDURES

### Preamble

In order to operate over areas of significant terrain, where descent to 10,000 ft may not be immediately possible, special depressurization procedures (or 'escape routes') have been created to meet the requirements of the Bahrain CAA ANTRs.

The escape routes published are predicated on the worst case scenario of full passenger capacity and a depressurization at the critical point (Decision Point). A conservative speed of 390kt TAS is assumed, together with complete exhaustion of the passenger oxygen system at 22 minutes. Descent to 14,000ft, or below, will be accomplished within this time period.

Descent profiles are derived from FCOM Procedures, Special Operations PRO SPO-20 and PRO SPO-35.

Further flight above 10,000ft but not above 14,000ft is permitted for a period not exceeding 30 min, after which supplementary oxygen must be provided for a minimum of 10% of the total passengers on board for the entire duration at these levels.

# Aircraft Performance

## DECOMPRESSION PROCEDURES

### Assumptions

Emergency descent is assumed to be with GEAR UP at Mmo/Vmo, aircraft structural considerations permitting.

En route descent between safety heights is assumed to be at 2,000ft / min. Two-minute allowance has been included for turns when calculating the maximum distance possible, and if a turn of 180° is required, re-establish on track as soon as possible.

Assumed average TAS of 390 kt.

Crew may divert to any suitable airport along the diversion route depending on the weather conditions prevailing.

Minimum Rescue Fire Fighting (RFF) category of CAT 4 is considered in the selection of diversion airports.

# Aircraft Performance

## DECOMPRESSION PROCEDURES

### Obstacle Clearance Policy

Obstacles clearance is applicable for:

Flying on airways.

Flying off airways.

Flying direct segments.

The MSA is calculated by considering all obstacles within a fifteen nautical mile (15 nm) buffer each side of the track, and a 15 nm radius at the end fixes, except when diverting to an airport where the MSA covers 15 nm either side of the preferred arrival route.

Obstacle data is extracted using the Grid MORA data provided by Jeppesen. However, in areas where Grid MORA is limiting, more precise obstacles data are extracted from topographical charts, a buffer of 2,000ft is added to the obstacle height, which has been rounded up to the next 1000ft.

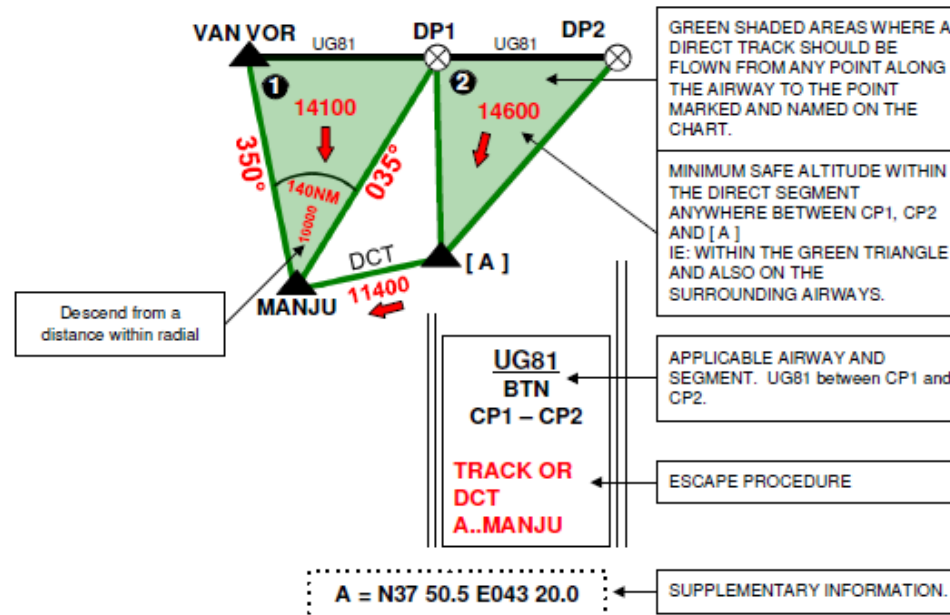
# Aircraft Performance

## DEPRESSURISATION PROCEDURES CHART LEGEND

**CAUTION:** Charts are not to scale.



|                   |                                                                 |
|-------------------|-----------------------------------------------------------------|
| 11700             | MSA.                                                            |
| →                 | Directional Arrow                                               |
| ↔                 | Bi-directional Arrow                                            |
| <b>RABEM</b><br>▲ | Waypoint with ident                                             |
| ⊙                 | Airport                                                         |
| UG81              | Airway ident                                                    |
| ⊗                 | Decision Point.<br>May also coincide with a Navaid or waypoint. |
| [ A ]             | FMGC waypoint ident for Critical Point.                         |
| —                 | Airway                                                          |
| —                 | Direct Segment - DCT                                            |
|                   | Identifies route segment where the diversion procedure applies. |
| ① ②               | Indicates Segment Number                                        |
|                   | Descend from distance within a radial                           |



ALL CRITICAL POINTS (DECISION POINTS) SHOWN ON CHARTS ARE FOR ZERO WIND AND MUST BE CORRECTED FOR WIND AT CRUISE LEVEL.

Medium Aircraft – MOVE CRITICAL POINTS UPWIND BY 2 nm per 10 kt of WIND  
Heavy Aircraft – MOVE CRITICAL POINTS UPWIND BY 3 nm per 10 kt of WIND

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Thank you.

